MID-TERM EXAMINATION

(Course Name : B. Tech. CSE/ECE/IT/MAE/DMAM) (Semester : 2nd)

(March, 2024) OFF LINE mode

Subject Code: BAS 102	**Subject: Applied Mathematics -II**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: Q. 1 is compulsory.	

Q1		**(2.5*4)**	
a	(a) Find the equation of tangent plane to the surface $x^3 + y^3 + 3xyz = 3$ at the point $(1, 2, -1)$.	UNIT-1	
d	(b) Determine the constant $'a'$ so that the vector $\vec{f} = (x + 3y)\hat{\imath} + (y - 2z)\hat{\jmath} + (x + az)\hat{k}$ is solenoidal.	UNIT-1	
	(c) Let $y_1 = \sin x$, $y_2 = \cos x$ & $y_3 = e^x$ be the solutions of homogeneous differential equation. Using the concept of Wronskian, check whether these solutions are linearly independent or dependent.	UNIT-2	
b	(d) Find out the particular integral of $(D^2 + 6D + 9)y = log5 + 4^x$.	UNIT-2	
Q2	**(Attempt any Two Parts) UNIT-1**		**(5,5)**
	(a) Check whether $\vec{V} = (z^2 + 2x + 3y)\hat{\imath} + (3x + 2y + z)\hat{\jmath} + (y + 2zx)\hat{k}$ represent the conservative vector field or not. If so determine the scalar potential.		
	(b) Find the total work done in moving a particle in the force field $\vec{F} = (\sin y)\hat{\imath} + x(1 + \cos y)\hat{\jmath}$ along the circular path given by $x^2 + y^2 = a^2,\ z = 0.$		
	(c) Verify Green's theorem for $\oint_C [(y - \sin x)\,dx + \cos x\ dy]$ where C is triangle enclosed by $y = 0,\ x = \frac{\pi}{2}, y = \frac{2x}{\pi}$.		
Q3	**(Attempt any Two Parts) UNIT-2**		**(5,5)**
	(a)Solve $(D^2 - 4D + 4)y = (1 + x)^2 e^{2x}$.		
	(b)Solve $(x^2D^2 - xD + 1)y = \left(\frac{logx}{x}\right)^2$.		
	(c) Using the method of undetermined coefficients, solve the differential equation $(D^2 - 2D + 3)y = e^x \sin x$.		